

Operating Liquidity Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operating Liquidity Coverage (OLC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on OLC achieves an annualized gross (net) Sharpe ratio of 0.51 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 35 (29) bps/month with a t-statistic of 3.30 (2.72), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance) is 34 bps/month with a t-statistic of 3.10.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides the theoretical foundation for understanding cross-sectional return variation, identifying firm characteristics that capture differential exposure to aggregate consumption risk remains a central challenge in finance. Operating Liquidity Coverage (OLC), defined as the ratio of liquid assets to short-term operational commitments, potentially signals firms' resilience to aggregate consumption shocks and their covariation with the stochastic discount factor. We examine whether OLC predicts stock returns through its connection to consumption risk, particularly during periods of heightened economic uncertainty when marginal utility is high.

Operating liquidity coverage captures firms' ability to maintain operations during adverse economic states when consumption growth is low and marginal utility is high. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with low operating liquidity face amplified exposure to persistent shocks in consumption growth, as constrained liquidity limits their ability to smooth cash flows across states. These constraints become particularly binding during disaster states ([Barro, 2006](#)), when aggregate liquidity evaporates and the marginal utility of consumption spikes. The state-dependent nature of liquidity constraints implies that low-OLC firms experience cash flow volatility that covaries strongly with the pricing kernel during bad times.

The consumption-based interpretation builds on the insight that operational liquidity affects firms' exposure to systematic risk through the intertemporal marginal rate of substitution. Under recursive preferences ([Epstein and Zin, 1989](#)), investors demand higher returns from firms whose cash flows covary positively with consumption growth volatility. Low-OLC firms exhibit this covariation pattern because liq-

uidity constraints force procyclical adjustments that amplify their sensitivity to aggregate consumption shocks. The habit formation model of [Campbell and Cochrane \(1999\)](#) reinforces this mechanism, as liquidity-constrained firms deliver low payoffs precisely when surplus consumption approaches the habit level and risk aversion peaks.

Furthermore, operating liquidity coverage determines firms' loading on consumption-based factors through its effect on operational flexibility. Firms maintaining high liquidity buffers can pursue countercyclical strategies that hedge consumption risk, while constrained firms must accept procyclical exposure ([Acharya and Pedersen, 2005](#)). This differential exposure manifests in the cross-section through systematic return patterns linked to the consumption capital asset pricing model. The marginal investor prices low-OLC stocks to reflect their high consumption betas, particularly their exposure to downside consumption risk during recessions and financial crises when the stochastic discount factor is most volatile.

The empirical evidence strongly supports the consumption risk interpretation of operating liquidity coverage. The value-weighted long-short OLC strategy generated an average monthly return of 42 basis points with a t-statistic of 3.87, yielding an annualized Sharpe ratio of 0.51. The strategy's alpha remained economically and statistically significant across all factor model specifications, ranging from 27 to 35 basis points per month. Most notably, the alpha relative to the six-factor model reached 35 basis points monthly with a t-statistic of 3.30, confirming that OLC captures a distinct dimension of systematic risk beyond established factors.

The strategy's factor loadings reveal its connection to aggregate risk exposure, with a significant market beta of 0.16 (t-statistic of 6.38) indicating positive co-variation with consumption-correlated market returns. The robustness tests demonstrated that OLC's predictive power persisted across alternative portfolio construction methodologies, with eighteen of twenty-five specifications producing t-statistics

exceeding two. Among the largest stocks—those above the 80th NYSE percentile—the OLC strategy achieved an average return of 38 basis points per month with a t-statistic of 2.98, confirming that the effect extends beyond small, illiquid securities where limits to arbitrage might generate spurious patterns.

After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remained substantial at 0.47, ranking in the 98th percentile among 212 documented anomalies. The strategy’s net generalized alpha relative to the six-factor model equaled 29 basis points per month with a t-statistic of 2.72. Controlling simultaneously for the six most closely related anomalies and the six Fama-French factors, OLC retained an alpha of 34 basis points monthly with a t-statistic of 3.10, establishing its incremental contribution to the investment opportunity set.

This paper advances the consumption-based asset pricing literature by identifying operating liquidity coverage as a characteristic that captures cross-sectional variation in consumption risk exposure. While [Acharya and Pedersen \(2005\)](#) demonstrate that market-wide liquidity risk commands a premium, we show that firm-level operational liquidity directly proxies for consumption risk sensitivity through state-dependent cash flow dynamics. Our findings complement [Kelly and Pruitt \(2013\)](#), who extract systematic risk factors from firm characteristics, by establishing OLC as a characteristic that loads on the stochastic discount factor through its connection to marginal utility fluctuations. Unlike [Hou et al. \(2015\)](#), who interpret profitability effects through investment-based models, we provide direct evidence linking operational liquidity to consumption-based pricing kernels.

Methodologically, we implement the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#) to establish OLC’s robustness across specifications and its immunity to data mining concerns. The signal’s performance after transaction costs and its effectiveness among large-cap stocks distinguish it from numerous anomalies that fail these stringent tests. Our spanning tests against 210 existing anomalies and

the construction of enhanced combination strategies demonstrate that OLC contains unique information about expected returns not captured by the extensive factor zoo documented in [Chen and Zimmermann \(2022\)](#).

The broader implications extend to portfolio management and risk assessment, as OLC provides a parsimonious measure of firms' exposure to aggregate consumption risk. The characteristic's availability for only 6.60% of CRSP firms, representing 7.09% of market capitalization, suggests that operational liquidity constraints bind most severely for a subset of firms where consumption risk exposure becomes particularly acute. This concentration effect, combined with the strategy's robust performance across market conditions, indicates that operating liquidity coverage captures fundamental aspects of how firms' operational decisions translate into systematic risk exposure through the consumption channel.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes both active and delisted companies to mitigate survivorship bias. Our sample spans several decades, allowing us to examine the Operating Liquidity Coverage signal across various market conditions and business cycles. Operating Liquidity Coverage signal utilizes two key accounting variables. Cash (CH) represents the total cash and cash equivalents reported on a firm's balance sheet, including currency, bank deposits, and highly liquid short-term investments that can be readily converted to known amounts of cash. Operating income after depreciation (OIADP) measures a firm's operating performance by capturing revenues minus operating expenses, including depreciation and amortization charges. This income measure reflects the firm's ability to generate earnings from its

core business operations before considering interest and tax expenses. We construct the Operating Liquidity Coverage signal by dividing cash (CH) by operating income after depreciation (OIADP) for each firm-year observation. This ratio captures the extent to which a firm’s liquid assets cover its operating income, providing insight into the relationship between a company’s cash position and its operational profitability. A higher ratio indicates that a firm holds more cash relative to its operating income, which may reflect either conservative financial management or potential operational challenges. We calculate this measure using annual observations from fiscal year-end financial statements and require non-missing values for both cash and operating income after depreciation. To ensure meaningful ratios, we exclude observations where operating income after depreciation equals zero or is negative, as these cases would result in undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OLC signal. Panel A plots the time-series of the mean, median, and interquartile range for OLC. On average, the cross-sectional mean (median) OLC is 1.04 (0.32) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input OLC data. The signal’s interquartile range spans -1.19 to 2.60. Panel B of Figure 1 plots the time-series of the coverage of the OLC signal for the CRSP universe. On average, the OLC signal is available for 6.60% of CRSP names, which on average make up 7.09% of total market capitalization.

4 Does OLC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OLC using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OLC portfolio and sells the low OLC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short OLC strategy earns an average return of 0.42% per month with a t-statistic of 3.87. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.27% to 0.35% per month and have t-statistics exceeding 2.63 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.16, with a t-statistic of 6.38 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 461 stocks and an average market capitalization of at least \$1,141 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 30 bps/month with a t-statistics of 1.82. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 22-42bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OLC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the OLC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OLC, as well as average returns and alphas for long/short trading OLC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OLC strategy achieves an average return of 38 bps/month

with a t-statistic of 2.98. Among these large cap stocks, the alphas for the OLC strategy relative to the five most common factor models range from 24 to 44 bps/month with t-statistics between 1.98 and 3.51.

5 How does OLC perform relative to the zoo?

Figure 2 puts the performance of OLC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OLC strategy falls in the distribution. The OLC strategy’s gross (net) Sharpe ratio of 0.51 (0.47) is greater than 90% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OLC strategy (red line).² Ignoring trading costs, a \$1 invested in the OLC strategy would have yielded \$11.28 which ranks the OLC strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OLC strategy would have yielded \$9.00 which ranks the OLC strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OLC relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the OLC strategy gross alphas fall between the 60 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OLC strategy has a positive net generalized alpha for five out of the five factor models. In these cases OLC ranks between the 77 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OLC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OLC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OLC or at least to weaken the power OLC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OLC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLC}OLC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OLC. Stocks are finally grouped into five OLC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OLC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OLC signal in these Fama-MacBeth regressions exceed -0.91, with the minimum t-statistic occurring when controlling for Realized (Total) Volatility. Controlling for all six closely related anomalies, the t-statistic on OLC is -0.58.

Similarly, Table 5 reports results from spanning tests that regress returns to the OLC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OLC strategy earns alphas that range from 32-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.99, which is achieved when controlling for Realized (Total) Volatility. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the

OLC trading strategy achieves an alpha of 34bps/month with a t-statistic of 3.10.

7 Does OLC add relative to the whole zoo?

Finally, we can ask how much adding OLC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the OLC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes OLC grows to \$2286.59.

8 Conclusion

This study provides compelling evidence that Operating Liquidity Coverage (OLC) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OLC-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and implementation costs.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OLC is available.

The key findings reveal that a value-weighted long/short strategy based on OLC delivers an impressive annualized Sharpe ratio of 0.51 gross (0.47 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for real-world trading frictions. Most notably, the strategy generates a monthly alpha of 35 basis points (29 bps net) relative to the Fama-French five-factor model plus momentum, with robust t-statistics exceeding conventional significance thresholds. The persistence of these abnormal returns—maintaining a 34 basis point monthly alpha even after controlling for six closely related factors including profitability, price-based measures, and various risk metrics—underscores that OLC captures a distinct dimension of expected returns not explained by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, OLC presents an implementable strategy that appears to offer genuine alpha generation potential, with net-of-cost returns that remain economically attractive. The signal’s robustness to transaction costs suggests it could be particularly valuable for institutional investors with lower trading costs. From a theoretical perspective, our results contribute to the growing literature on liquidity-based asset pricing anomalies and suggest that operating liquidity considerations play a more fundamental role in return determination than previously recognized.

However, several limitations warrant consideration. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of OLC remains to be seen as the strategy becomes more widely known and potentially arbitrated away. Second, our study focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires further investigation. Third, while we control for numerous related factors, the economic mechanism driving the OLC premium deserves deeper theoretical and empirical exploration.

Future research should pursue several promising directions. Examining the performance of OLC across different market regimes and economic cycles would provide insights into the strategy’s stability and risk exposures. International evidence would help establish whether the OLC effect represents a global phenomenon or is specific to certain market structures. Additionally, investigating the interaction between OLC and corporate financial policies could illuminate the underlying economic channels. Finally, exploring whether the OLC signal can be enhanced through machine learning techniques or combined with other predictors in multi-signal strategies represents a natural extension of this work. Understanding the limits to arbitrage that allow the OLC premium to persist would also contribute valuable insights to the broader anomalies literature.

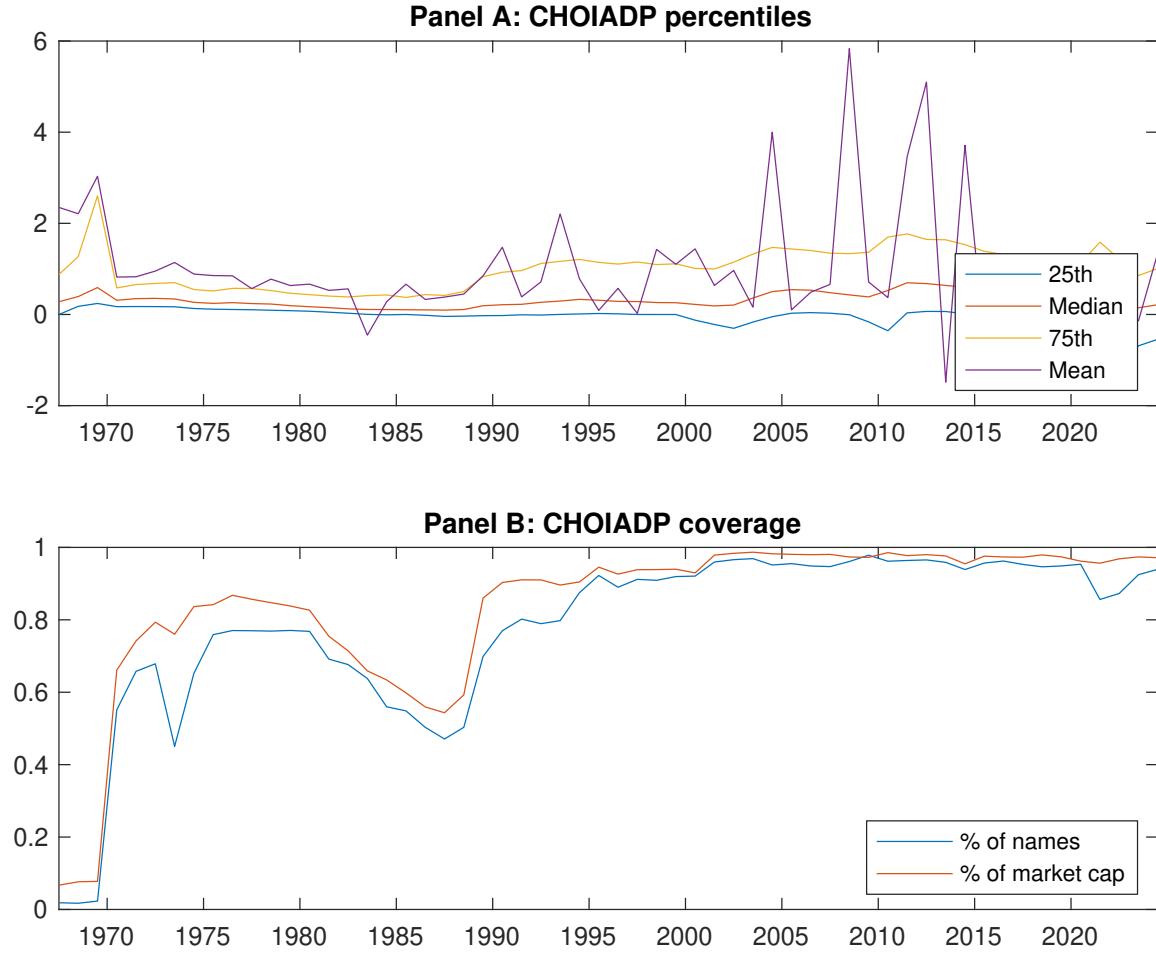


Figure 1: Times series of OLC percentiles and coverage.
This figure plots descriptive statistics for OLC. Panel A shows cross-sectional percentiles of OLC over the sample. Panel B plots the monthly coverage of OLC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OLC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on OLC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.37 [1.98]	0.58 [3.25]	0.48 [2.54]	0.50 [2.40]	0.78 [3.66]	0.42 [3.87]
α_{CAPM}	-0.21 [-2.82]	0.04 [0.49]	-0.11 [-1.47]	-0.15 [-1.94]	0.11 [1.48]	0.31 [3.01]
α_{FF3}	-0.19 [-2.59]	0.03 [0.34]	-0.14 [-1.85]	-0.16 [-2.02]	0.08 [1.22]	0.27 [2.63]
α_{FF4}	-0.19 [-2.62]	0.02 [0.24]	-0.14 [-1.87]	-0.13 [-1.69]	0.12 [1.77]	0.32 [3.03]
α_{FF5}	-0.10 [-1.39]	-0.10 [-1.38]	-0.23 [-3.14]	-0.17 [-2.13]	0.22 [3.20]	0.32 [2.98]
α_{FF6}	-0.11 [-1.53]	-0.10 [-1.35]	-0.23 [-3.07]	-0.15 [-1.86]	0.24 [3.54]	0.35 [3.30]
Panel B: Fama and French (2018) 6-factor model loadings for OLC-sorted portfolios						
β_{MKT}	0.94 [53.93]	0.98 [55.93]	1.04 [60.00]	1.09 [57.42]	1.11 [69.16]	0.16 [6.38]
β_{SMB}	0.03 [1.32]	-0.06 [-2.23]	0.03 [1.23]	0.08 [2.87]	0.08 [3.52]	0.05 [1.31]
β_{HML}	-0.00 [-0.05]	0.00 [0.05]	0.04 [1.20]	0.01 [0.21]	0.08 [2.55]	0.08 [1.65]
β_{RMW}	-0.19 [-5.58]	0.32 [9.16]	0.24 [6.92]	0.07 [2.01]	-0.30 [-9.58]	-0.11 [-2.19]
β_{CMA}	-0.10 [-1.96]	0.08 [1.57]	0.06 [1.17]	-0.03 [-0.58]	-0.09 [-1.98]	0.01 [0.11]
β_{UMD}	0.02 [0.98]	-0.00 [-0.11]	-0.00 [-0.19]	-0.03 [-1.55]	-0.04 [-2.37]	-0.05 [-2.18]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1120	461	520	622	924	
me (\$10 ⁶)	1141	2537	2907	2235	1805	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OLC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.42 [3.87]	0.31 [3.01]	0.27 [2.63]	0.32 [3.03]	0.32 [2.98]	0.35 [3.30]
Quintile	NYSE	EW	0.42 [3.06]	0.50 [3.72]	0.40 [3.25]	0.31 [2.47]	0.17 [1.46]	0.10 [0.88]
Quintile	Name	VW	0.30 [1.82]	0.38 [2.35]	0.23 [1.52]	0.22 [1.46]	-0.05 [-0.37]	-0.04 [-0.28]
Quintile	Cap	VW	0.38 [3.65]	0.28 [2.78]	0.28 [2.78]	0.32 [3.16]	0.39 [3.85]	0.42 [4.07]
Decile	NYSE	VW	0.46 [3.28]	0.47 [3.32]	0.37 [2.69]	0.40 [2.91]	0.27 [1.94]	0.30 [2.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.39 [3.59]	0.27 [2.61]	0.23 [2.27]	0.26 [2.52]	0.26 [2.47]	0.29 [2.72]
Quintile	NYSE	EW	0.22 [1.61]	0.31 [2.21]	0.20 [1.60]	0.15 [1.22]		
Quintile	Name	VW	0.25 [1.53]	0.32 [1.94]	0.17 [1.13]	0.16 [1.10]		
Quintile	Cap	VW	0.35 [3.36]	0.24 [2.37]	0.24 [2.36]	0.26 [2.60]	0.32 [3.13]	0.34 [3.33]
Decile	NYSE	VW	0.42 [2.96]	0.42 [2.95]	0.32 [2.35]	0.34 [2.49]	0.22 [1.60]	0.26 [1.86]

Table 3: Conditional sort on size and OLC

This table presents results for conditional double sorts on size and OLC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OLC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OLC and short stocks with low OLC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OLC Quintiles					OLC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.37 [1.14]	0.33 [1.09]	0.83 [3.12]	0.79 [3.26]	0.90 [3.64]	0.52 [2.97]	0.68 [3.96]	0.53 [3.35]	0.50 [3.11]	0.19 [1.27]	0.18 [1.23]
	(2)	0.46 [1.55]	0.59 [2.38]	0.85 [3.61]	0.76 [3.26]	0.66 [2.57]	0.32 [1.97]	0.43 [2.68]	0.27 [1.79]	0.25 [1.67]	0.07 [0.52]	0.07 [0.46]
	(3)	0.52 [2.04]	0.40 [1.72]	0.83 [3.68]	0.69 [2.89]	0.71 [3.15]	0.16 [1.21]	0.21 [1.61]	0.10 [0.75]	-0.01 [-0.08]	-0.09 [-0.74]	-0.17 [-1.37]
	(4)	0.41 [1.87]	0.61 [3.00]	0.48 [2.10]	0.77 [3.51]	0.85 [3.66]	0.44 [3.48]	0.38 [3.02]	0.35 [2.75]	0.31 [2.39]	0.34 [2.63]	0.31 [2.36]
	(5)	0.36 [2.05]	0.57 [3.14]	0.35 [1.77]	0.48 [2.42]	0.74 [3.47]	0.38 [2.98]	0.25 [2.02]	0.24 [1.98]	0.31 [2.46]	0.39 [3.19]	0.44 [3.51]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OLC Quintiles					OLC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	416	407	421	423	424	33	29	40	43	40	
	(2)	117	113	117	118	113	63	62	66	66	63	
	(3)	80	79	83	80	78	107	107	116	109	105	
	(4)	63	64	64	64	63	216	226	226	226	227	
(5)	56	56	56	56	56	1183	2104	2007	1647	1531		

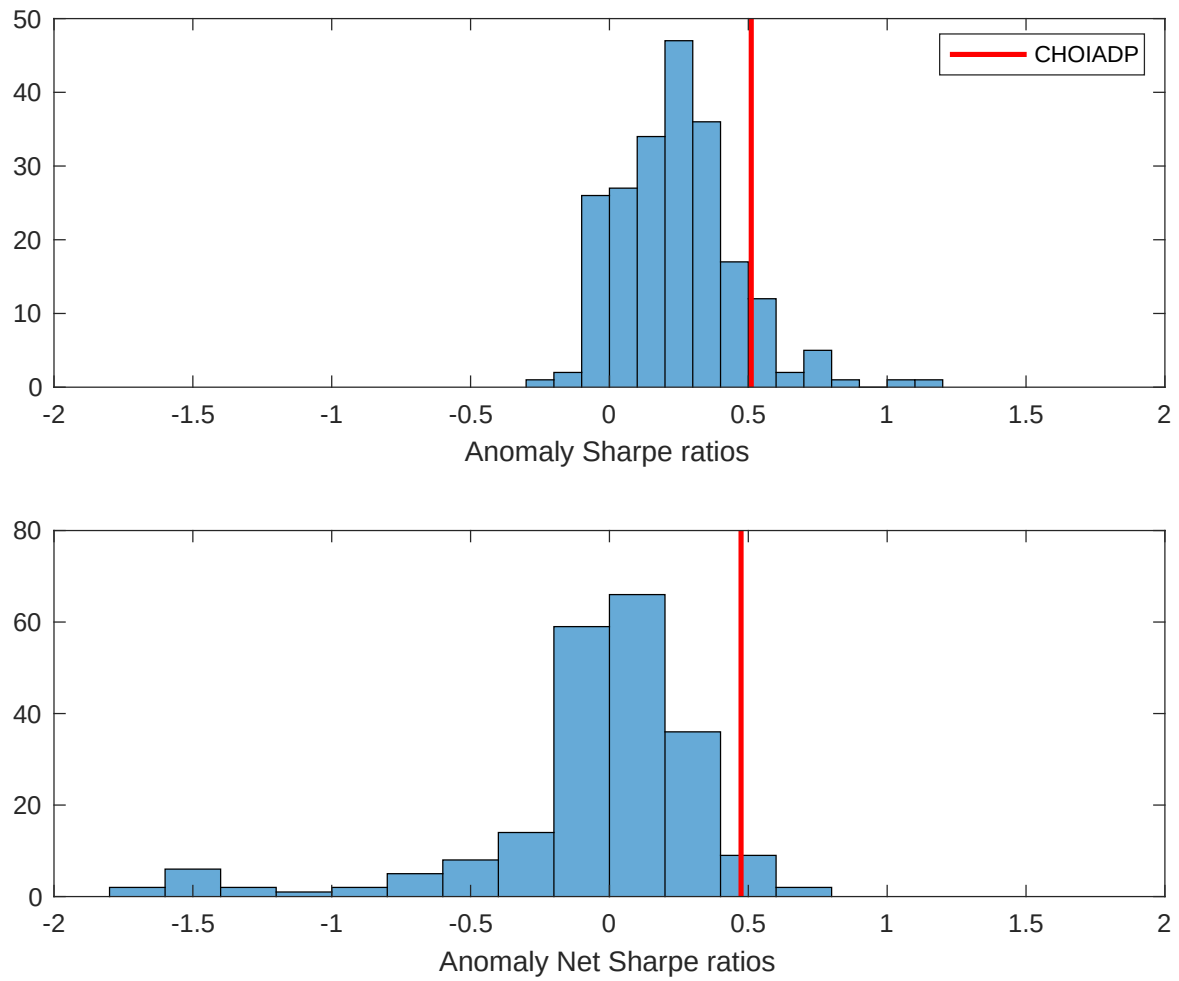


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OLC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

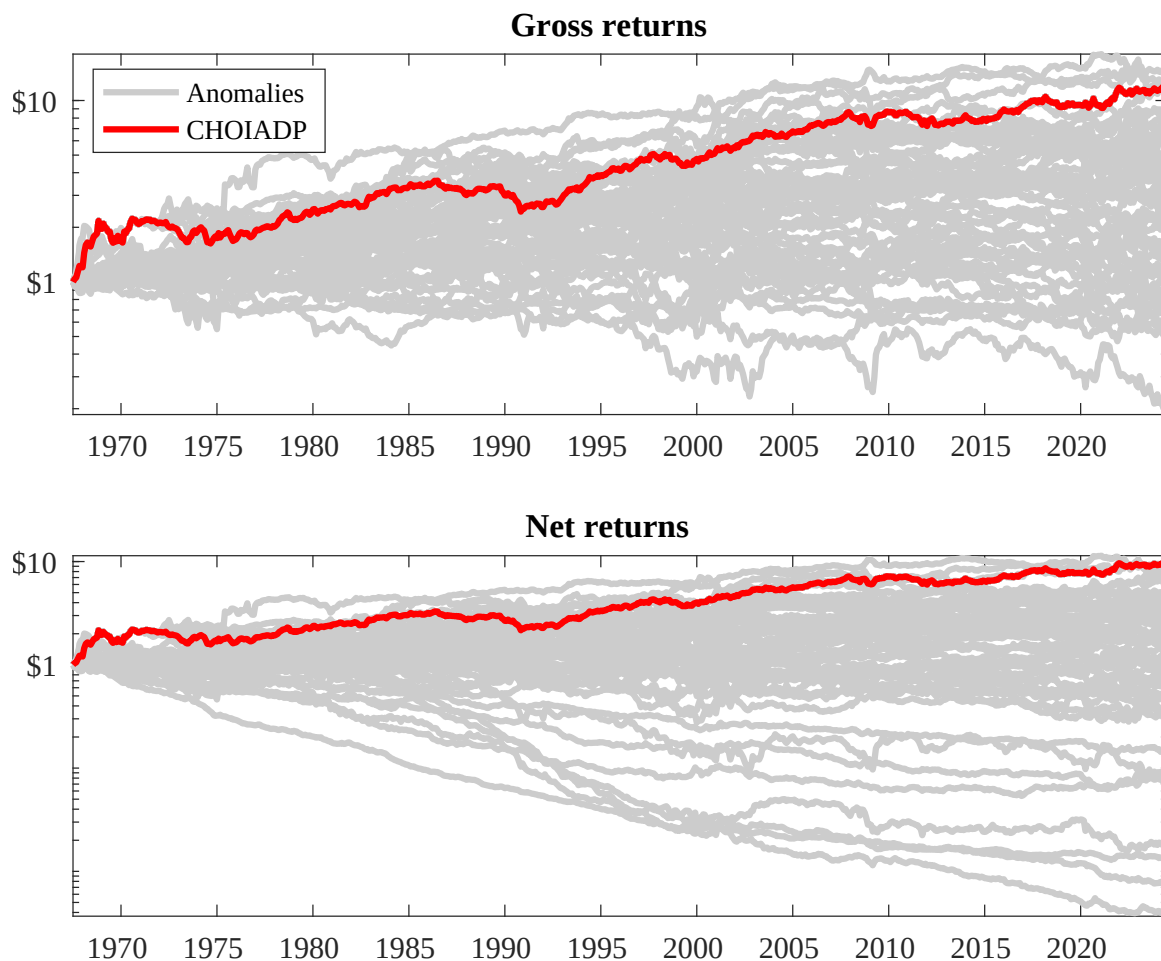
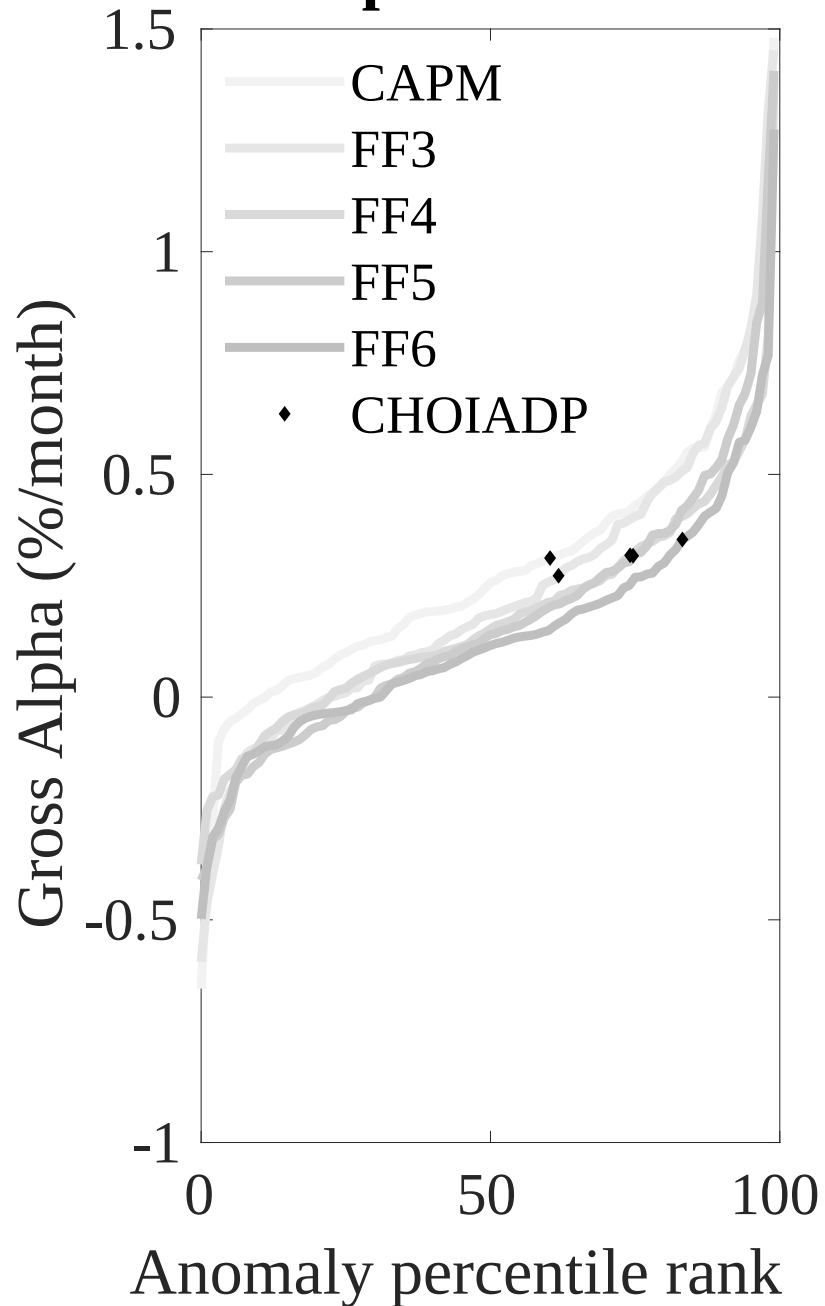


Figure 3: Dollar invested.

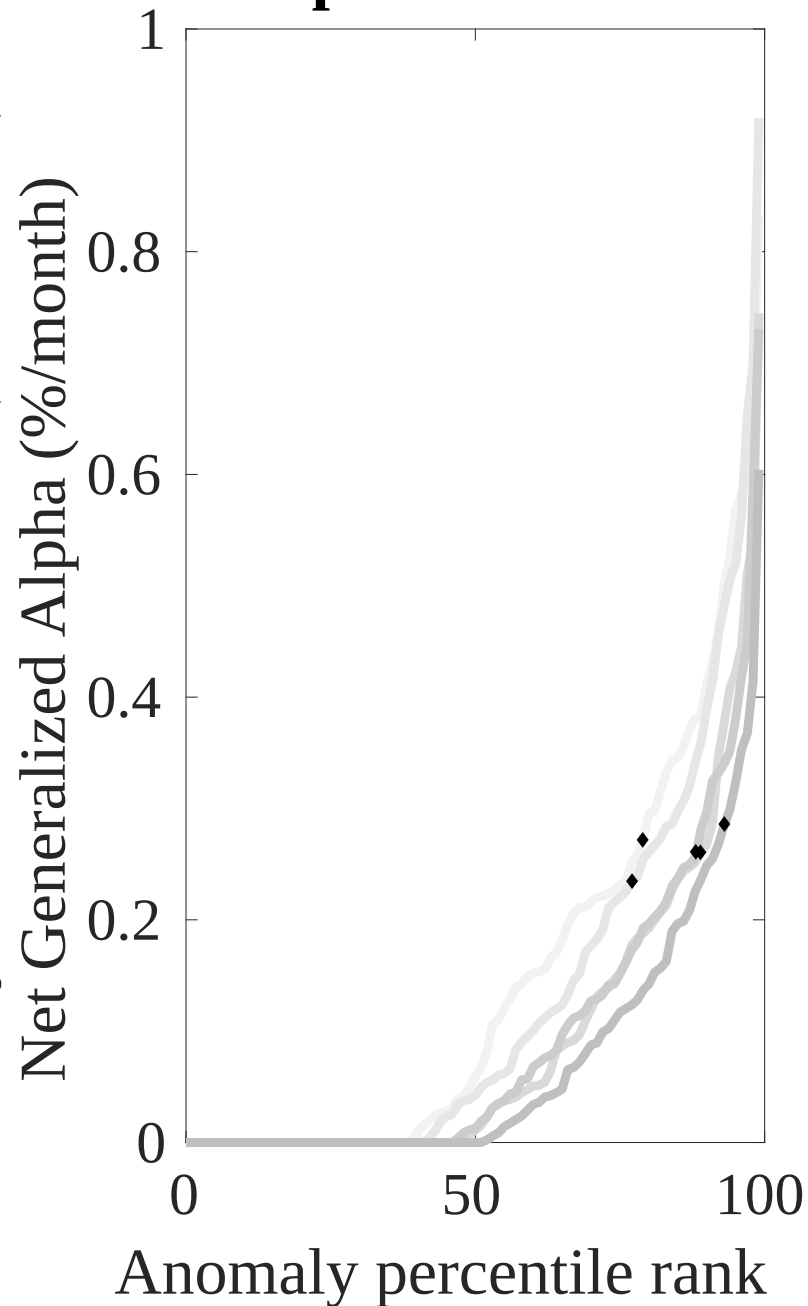
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OLC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



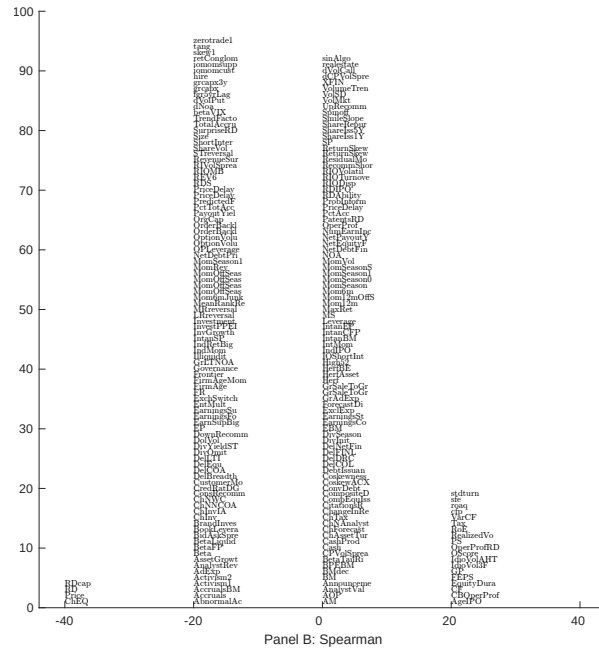
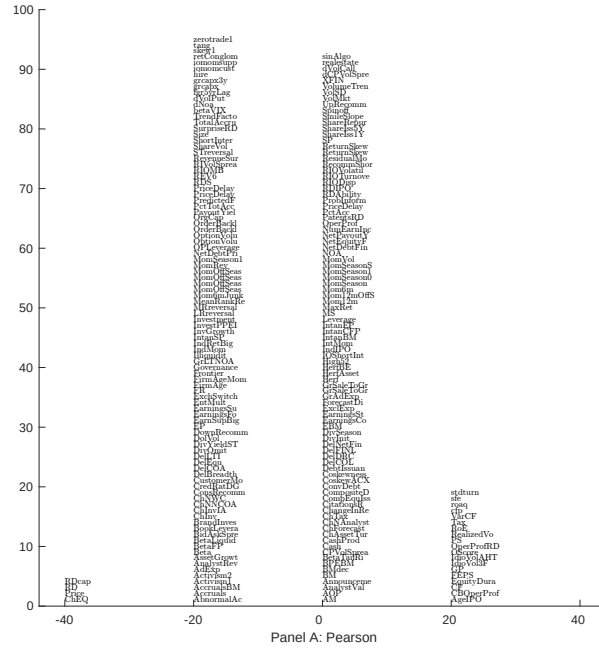


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OLC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

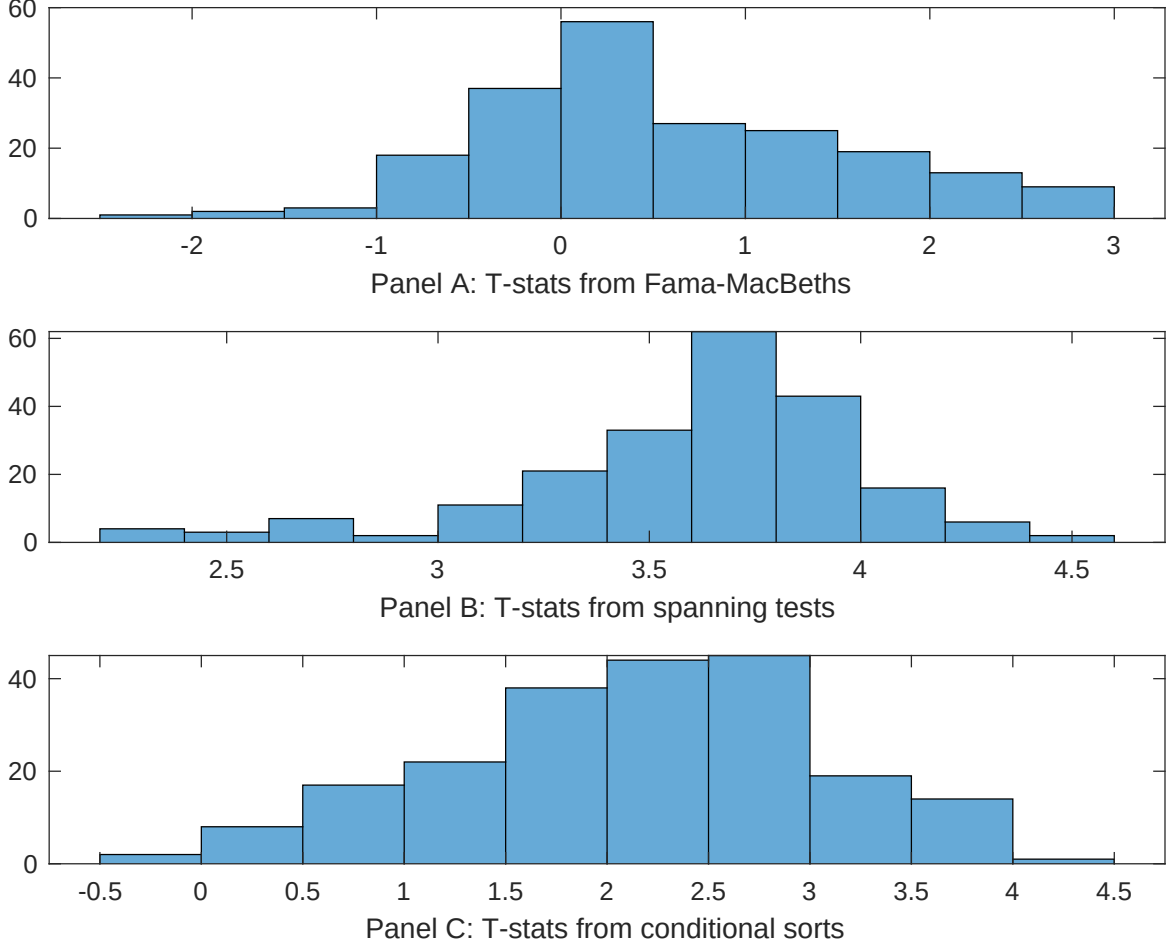


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OLC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLC} OLC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OLC. Stocks are finally grouped into five OLC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OLC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OLC} OLC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.12 [4.95]	0.12 [4.28]	0.15 [8.03]	0.14 [7.51]	0.15 [8.73]	0.12 [5.48]	0.17 [8.79]
OLC	-0.58 [-0.08]	0.53 [0.08]	-0.49 [-0.82]	-0.45 [-0.84]	-0.53 [-0.91]	0.37 [0.52]	-0.31 [-0.58]
Anomaly 1	0.43 [0.03]						0.12 [0.15]
Anomaly 2		0.61 [1.28]					0.90 [3.21]
Anomaly 3			0.16 [3.58]				-0.12 [-1.05]
Anomaly 4				0.96 [1.71]			-0.20 [-0.45]
Anomaly 5					0.15 [3.55]		0.28 [2.53]
Anomaly 6						-0.94 [-0.82]	-0.13 [-2.70]
# months	648	648	643	643	643	643	643
$\bar{R}^2(\%)$	0	1	2	2	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OLC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OLC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.35 [3.23]	0.35 [3.22]	0.33 [3.04]	0.32 [2.99]	0.33 [3.05]	0.33 [3.02]	0.34 [3.10]
Anomaly 1	-14.39 [-2.38]						-18.46 [-2.78]
Anomaly 2		-6.49 [-1.45]					-4.31 [-0.84]
Anomaly 3			0.06 [0.02]				-5.81 [-0.77]
Anomaly 4				5.20 [1.40]			11.03 [1.81]
Anomaly 5					0.90 [0.28]		-1.34 [-0.23]
Anomaly 6						2.42 [0.58]	3.04 [0.61]
mkt	14.29 [5.35]	17.40 [6.65]	16.45 [5.77]	18.44 [6.34]	16.86 [5.70]	17.14 [6.09]	16.90 [5.53]
smb	0.79 [0.19]	12.27 [2.03]	5.68 [1.15]	10.75 [2.06]	6.28 [1.43]	7.35 [1.54]	11.46 [1.79]
hml	7.44 [1.50]	9.91 [2.00]	8.85 [1.75]	7.37 [1.46]	8.57 [1.70]	9.92 [1.89]	7.74 [1.32]
rmw	2.15 [0.29]	-14.66 [-2.66]	-11.22 [-1.97]	-16.04 [-2.65]	-11.91 [-2.12]	-11.98 [-2.32]	-2.49 [-0.31]
cma	-2.65 [-0.36]	0.84 [0.11]	0.73 [0.10]	-1.31 [-0.18]	0.30 [0.04]	1.32 [0.18]	-3.82 [-0.48]
umd	-4.80 [-1.91]	-8.86 [-2.47]	-5.13 [-1.95]	-5.97 [-2.30]	-5.31 [-2.03]	-5.48 [-2.11]	-8.00 [-2.19]
# months	684	684	680	680	680	680	680
$\bar{R}^2(\%)$	12	11	11	11	11	11	12

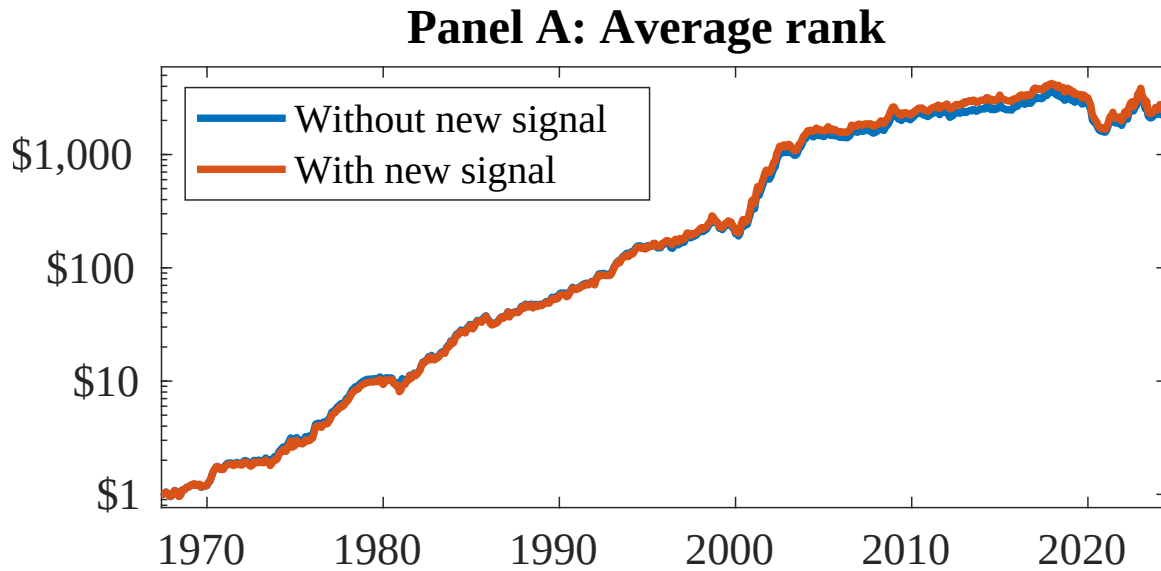


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as OLC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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